Stantec

CEMEX Lyons Access Traffic Study

Lyons Plant

August 28, 2023

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CEMEX LYONS ACCESS TRAFFIC STUDY

Revision	Description	Author	Date	Quality Check	Date	Independent Review	Date
A	Draft	BH	8/8/2023	VE	8/11/2023	AE	8/11/2023
B	Draft	BH	8/18/2023	VE	8/18/2023	AE	8/18/2023

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1.0 INTRODUCTION

The 2022 closure of CEMEX's Dowe Flats Quarry, located north of Ute Highway (SH 66), has impacted daily truck traffic, both entering and exiting the CEMEX cement plant, south of SH 66. Located in Lyons, Colorado and within Boulder County, the quarry previously provided a portion of the limestone and shale to CEMEX's cement plant via conveyor belt over SH 66, with the balance of the supply of these products to the plant being received by truck. With the Dowe Flats Quarry now closed, these products are now being brought in exclusively by truck. Stantec was tasked with evaluating the overall increase in truck traffic as a result of the quarry closure, and its impact to the adjacent roadway network.

1.1 STUDY AREA

The study area for this analysis included four intersections along SH 66, between N Foothills Highway (US 36) and N 53rd Street. These intersections are listed below and are illustrated in **Figure 1**.

1. US 36 & SH 66
2. SH 66 & Highland Drive (East Access)
3. SH 66 & CEMEX Driveway
4. SH 66 & N 53rd Street

Figure 1. CEMEX TIS Study Intersections



2.0 DATA COLLECTION

In order to define a baseline for traffic growth and future operations, existing conditions were assessed for the entirety of the project area. Because this study is being performed after the closure of the Dowe Flats Quarry, existing conditions have been defined as June 2022; three months prior to closure. By considering traffic operations during this time, it is possible to analyze a snapshot of what conditions were like before the study area started to experience an increase in truck traffic.

Data collected as part of this study included traffic counts, regional traffic forecasts, existing roadway geometry, and timestamped truck data logs. Further information on this data is described below.

- Traffic counts within the study area were obtained from All Traffic Data Services, Inc. and are included in **Appendix A**. The following types of counts were collected in June 2023:

  - **24-Hour Tube Counts:** Collected over three days (Tuesday, June 13th, Wednesday, June 14th, and Thursday, June 15th), these counts provide Average Daily Traffic (ADT) information for one location along SH 66, east of the CEMEX access driveway. These counts are collected in 15-minute intervals and summed to show the rolling peak hour trends over the course of each day. These counts are used to identify which single day to analyze as part of this study. **Figure 2** illustrates hourly volumes for each day that counts were collected, based on the raw data detailed in **Appendix A**. These counts do not separate out vehicle classification when summing the total ADT. Thursday, June 15th was selected for analysis purposes and can be considered representative of a typical weekday.

  - **Turning Movement Counts (TMCs):** Collected over the course of three weekdays (Tuesday, June 13th, Wednesday, June 14th, and Thursday, June 15th), these counts show the number of vehicles performing each movement (left-turn, through, or right-turn), on each approach and at all study intersections. These counts are collected in 15-minute intervals and summed to identify AM and PM peak hour volumes at each intersection. Passenger vehicles and heavy vehicles are identified separately within the counts. TMCs also include bicycle and pedestrian data at each location. Vehicular TMCs are displayed in **Figure 3**.

Figure 2. Daily Traffic – Collected Counts

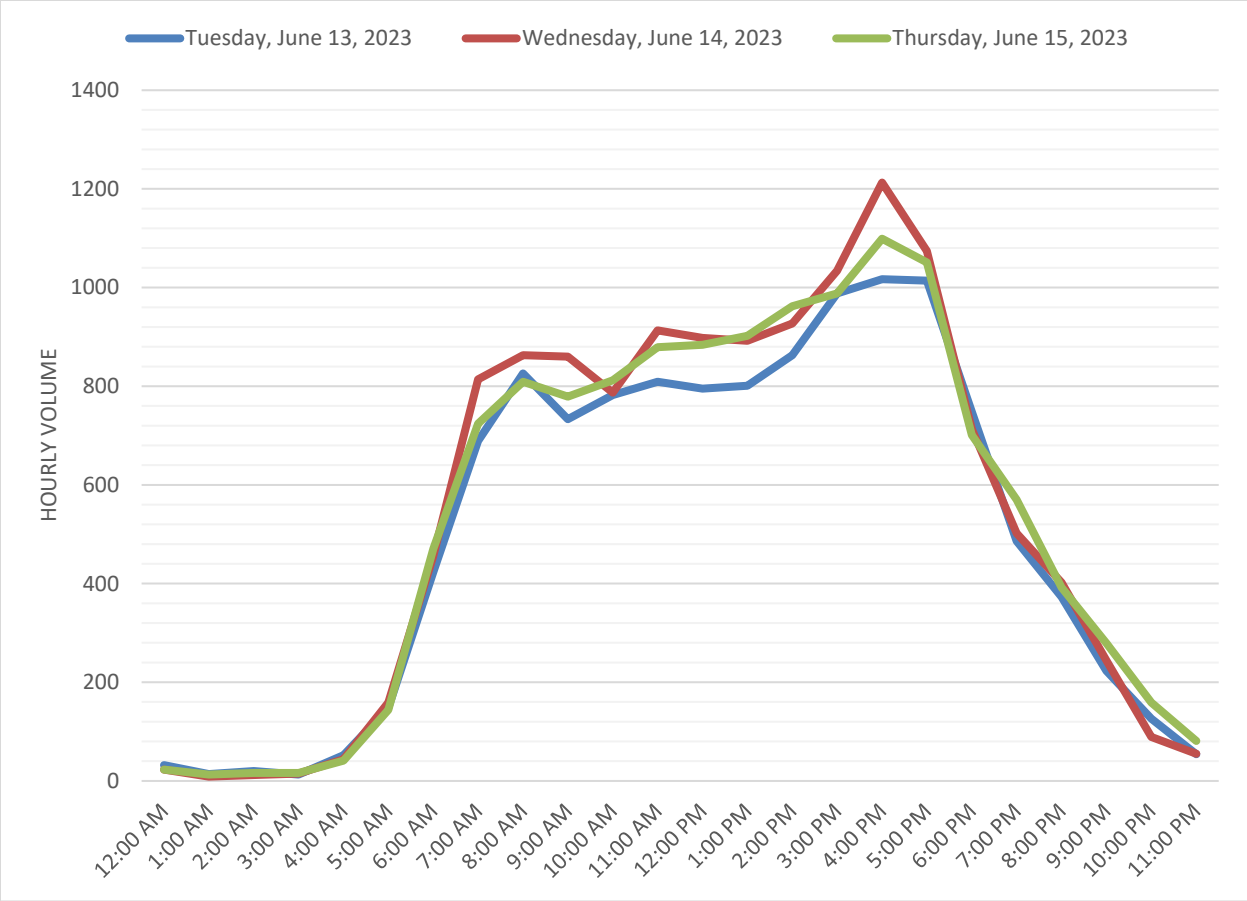
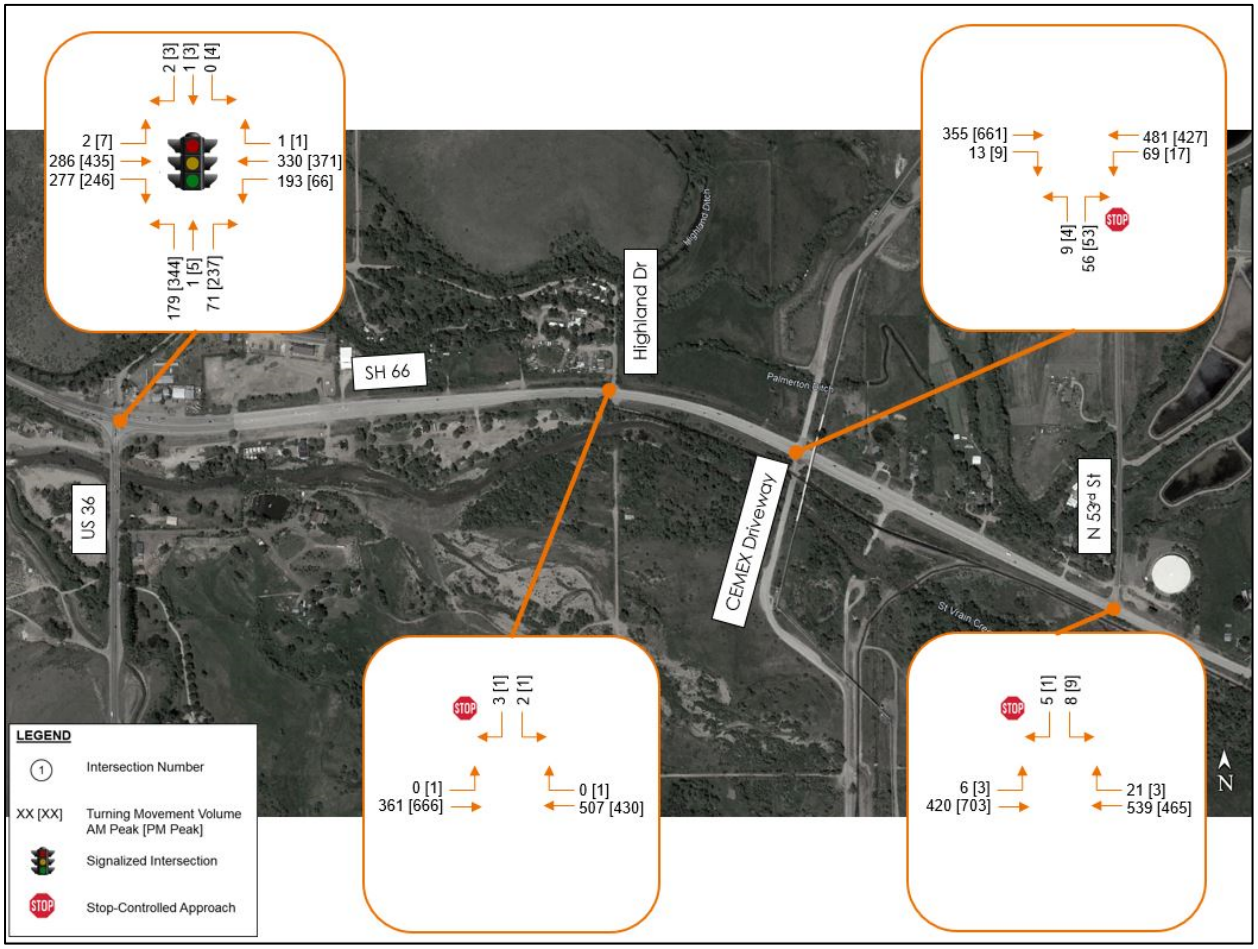


Figure 3. Turning Movement Counts – June 15, 2023



* All volumes reported as passenger car equivalent (PCE)

- For the purpose of developing growth rates in the vicinity of the CEMEX site, *Model Assigned All-Day Volumes* from the Denver Regional Council of Governments (DRCOG) were obtained for the years 2020 and 2050. The DRCOG travel model, "Focus", forecasts data related to roadway users, transit riders, and travel patterns. By incorporating data about people, businesses, transportation systems, and development patterns, daily volumes are calculated and made available in PDF form. **Table 1** shows these daily volumes within the study area. PDFs showing the DRCOG 2020 and 2050 outputs are included in **Appendix B**.

Table 1. DRCOG Model Assigned All Day Traffic Volumes

Location	SH 66 (US 36 to N 63rd St)	
Model Year	2020	2050
Daily Volume	10,000	15,000
CAGR	1.5%	

- Roadway geometry was obtained for the purpose of further understanding the existing network, as well as for creating Synchro models. This information was gathered using Google Earth and Bing Maps.
- Data logs provided by CEMEX contained a detailed breakdown of trucks entering the site between 2022 and 2023. Trips were broken down by specific date, time, and material. This data was sorted through to compare the number of truck trips before and after the Dowe Flats Quarry closure.

2.1 ROADWAY NETWORK

Geometry and roadway classifications were determined using a combination of Google Earth imagery and the *2018 Boulder County Road Map*. Characteristics of the existing roadway network within the study area are described below.

N Foothills Highway (US 36) is classified as a Principal Arterial roadway. US 36 is an east-west, Federal Highway that provides regional connectivity from the town of Estes Park to Denver, Interstate 25, and Interstate 270. From Boulder to Estes Park, US 36 is a two-lane rural highway with rolling grades and posted speed limits ranging from 50 to 60 mph.

Ute Highway (SH 66) connects the towns of Platteville, Longmont, and Lyons and is defined as a two-lane, rural, Principal Arterial. This east-west state highway has flat grades throughout and provides direct access to the CEMEX site. Posted speed limits range from 35 to 45 mph towards the eastern limits and 55 to 60 mph to the west.

Highland Drive is a rural, local roadway that loops from the US 36 intersection to SH 66, just west of the CEMEX site. Roadway material at this location is gravel. With a posted speed limit of 20 mph and flat grades, Highland Drive accesses residential properties.

CEMEX Driveway is the main access for the CEMEX cement plant. This is a two-lane private roadway. There are eastbound right-turn and westbound left-turn lanes on SH 66 to allow access to the site.

N 53rd Street is a two-lane, rural, local roadway located at the eastern limit of the study area. This roadway runs north-south and connects SH 66 to North Dakota Ridge Road and additional local street networks. The posted speed limit in the vicinity of the project site is 25 mph.

3.0 TRIP GENERATION

Trips generated by the site, both entering and exiting, were calculated for pre- and post-Dowe Flats Quarry closure scenarios. Trucks hauling limestone and shale are the primary reason for the increase in observed traffic entering and exiting the site. The change in the number of employees entering and exiting the site, when comparing 2022 to 2023, is negligible. Per the *State of Colorado State Highway Access Code*, all trips discussed in this study are reported as passenger car equivalents, converting each truck to three passenger cars.

3.1 2023 (POST-DOWE FLATS CLOSURE)

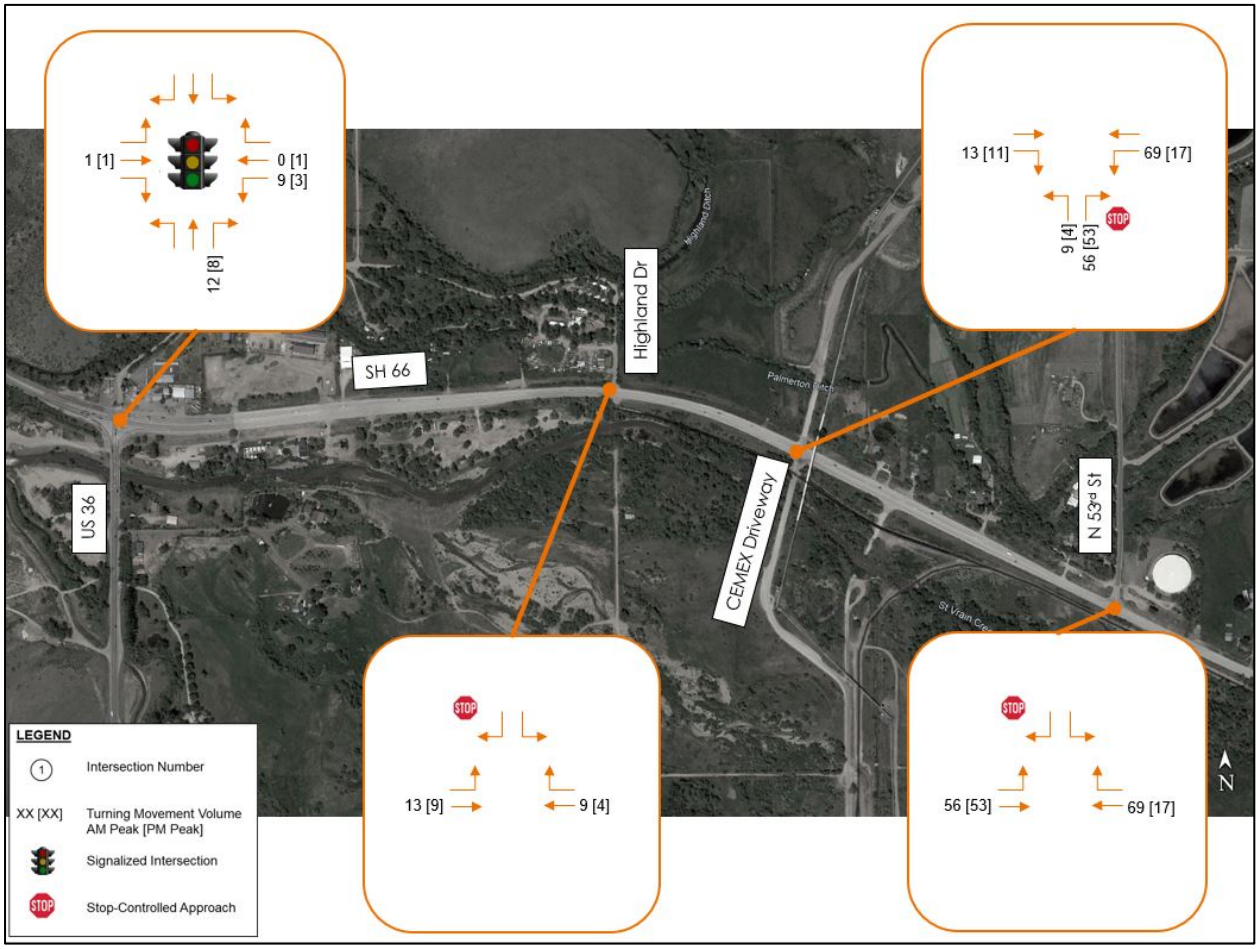
Turning movement counts, collected in June 2023, were used to determine the number of trips generated by the CEMEX cement plant after the Dowe Flats Quarry closure. The total of the turning volumes for the eastbound right and westbound left movements signified the number of trips entering the site daily and during the AM peak hour and PM peak hour. Similarly, northbound left and northbound right movements were considered for the exiting trips. Entering and exiting trips were added together to obtain the total number of trips for a given time period. These trips are summarized in **Table 2**. Trips generated by the site on June 15th, 2023, are displayed in **Figure 4**.

Table 2. 2023 CEMEX Trip Generation Values

Location	Daily			AM (8:15-9:15)			PM (4:00-5:00)		
	Total	Entering	Exiting	Total	Entering	Exiting	Total	Entering	Exiting
CEMEX Cement Plant	1283	642	641	147	82	65	85	28	57

* All volumes reported as passenger car equivalent (PCE)

Figure 4. CEMEX Trip Generation – June 15, 2023



* All volumes reported as passenger car equivalent (PCE)

Of the total number of trips generated by the CEMEX cement plant, shown above, data logs provided by CEMEX showed that on June 15th, 2023, there were 132 daily, 14 AM peak hour and 8 PM peak hour limestone and shale trips entering the site.

3.2 2022 (PRE-DOWE FLATS CLOSURE)

Trips generated by the CEMEX cement plant, prior to the Dowe Flats closure, were used to determine the 2023 increase in truck traffic while also obtaining a baseline for cement plant operations prior the closure of the quarry. To be consistent with 2023 traffic counts and seasonality in the vicinity of the site, material hauling data from June 2022 was used for this trip generation analysis.

CEMEX data logs were utilized to calculate the average number of truck trips for total daily, AM peak, and PM peak conditions. Using June 2022 as the sample size, there was an average of 17 daily, 3 AM peak

hour, and 1 PM peak hour inbound limestone and shale truck trips. These values were then subtracted from the 2023 CEMEX logged limestone and shale truck trips to determine the approximate increase in truck trips as a result of the quarry closure. Knowing these differences, the 2023 trip generation values were reduced accordingly to estimate the 2022 trip generation for all vehicles entering and exiting the CEMEX cement plant, shown in **Table 3** and **Figure 5**.

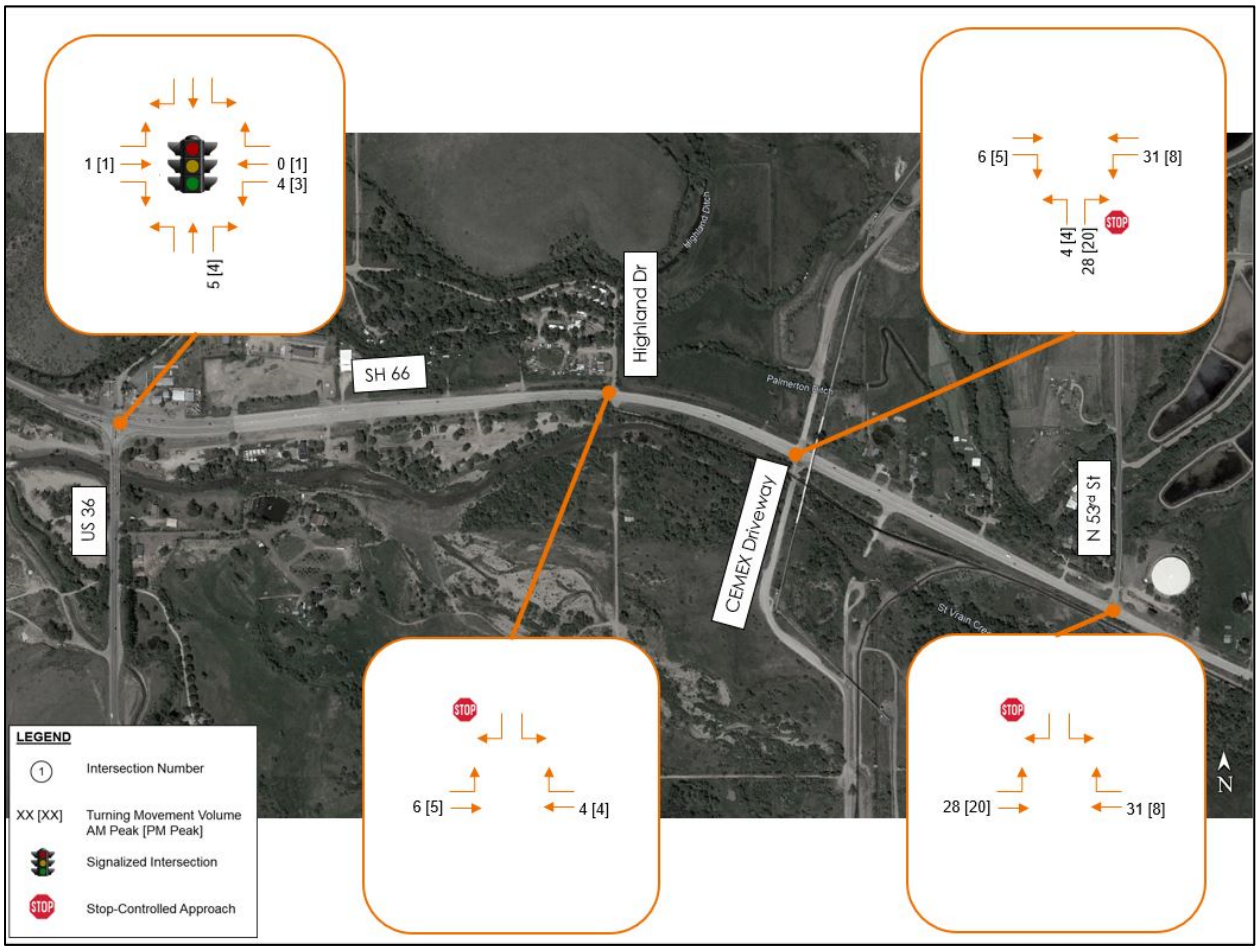
Trip distribution was determined using a combination of 2023 distribution, determined from 2023 TMCs, and directional information provided by CEMEX. It was assumed that the only change in trip generation due to the Dowe Flats Quarry closure is the number of limestone and shale trucks entering and exiting the site; the number of employees commuting to the site was assumed to remain the same. The increase in trip generation values from 2022 to 2023 is illustrated in **Figure 6**.

Table 3. 2022 Estimated CEMEX Trip Generation Values

Location	Daily			AM (8:15-9:15)			PM (4:00-5:00)		
	Total	Entering	Exiting	Total	Entering	Exiting	Total	Entering	Exiting
CEMEX Cement Plant	593	297	296	69	37	32	36	12	24

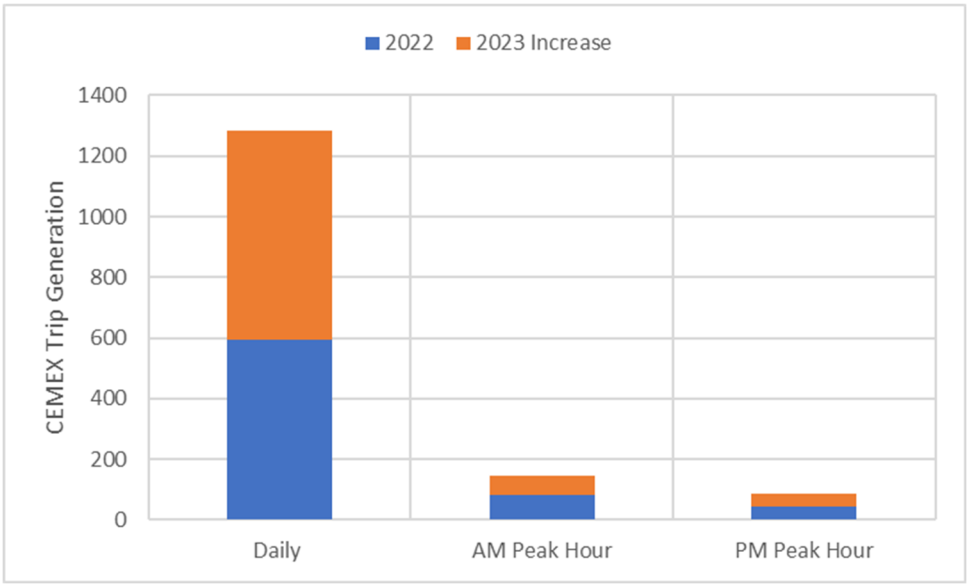
* All volumes reported as passenger car equivalent (PCE)

Figure 5. CEMEX Trip Generation – 2022 (Estimated)



* All volumes reported as passenger car equivalent (PCE)

Figure 6. CEMEX Trip Generation Increase – 2022 vs. 2023



* All volumes reported as passenger car equivalent (PCE)

3.3 MATERIAL HAULING ROUTES

An analysis of material hauling routes provided opportunities to further explore conditions of the roadways that trucks are using. 2023 existing traffic counts showed that the majority of material hauling trips utilize SH 66 east of the CEMEX cement plant. Existing westbound left-turn lanes provide storage for trucks entering the site. It is assumed that these vehicles access SH 66 via Interstate 25 to the east. Additionally, an eastbound acceleration lane is present for trucks exiting the cement plant to the east.

While the majority of trucks access the cement plant from the east, per CEMEX staff, inbound shale trucks access the cement plant from the west, traveling northbound on US 36 and eastbound on SH 66. Currently there is a channelized northbound right-turn and exclusive westbound left-turn lane present at the intersection of US 36 and SH 66 to accommodate these movements.

4.0 SYNCHRO MODELS

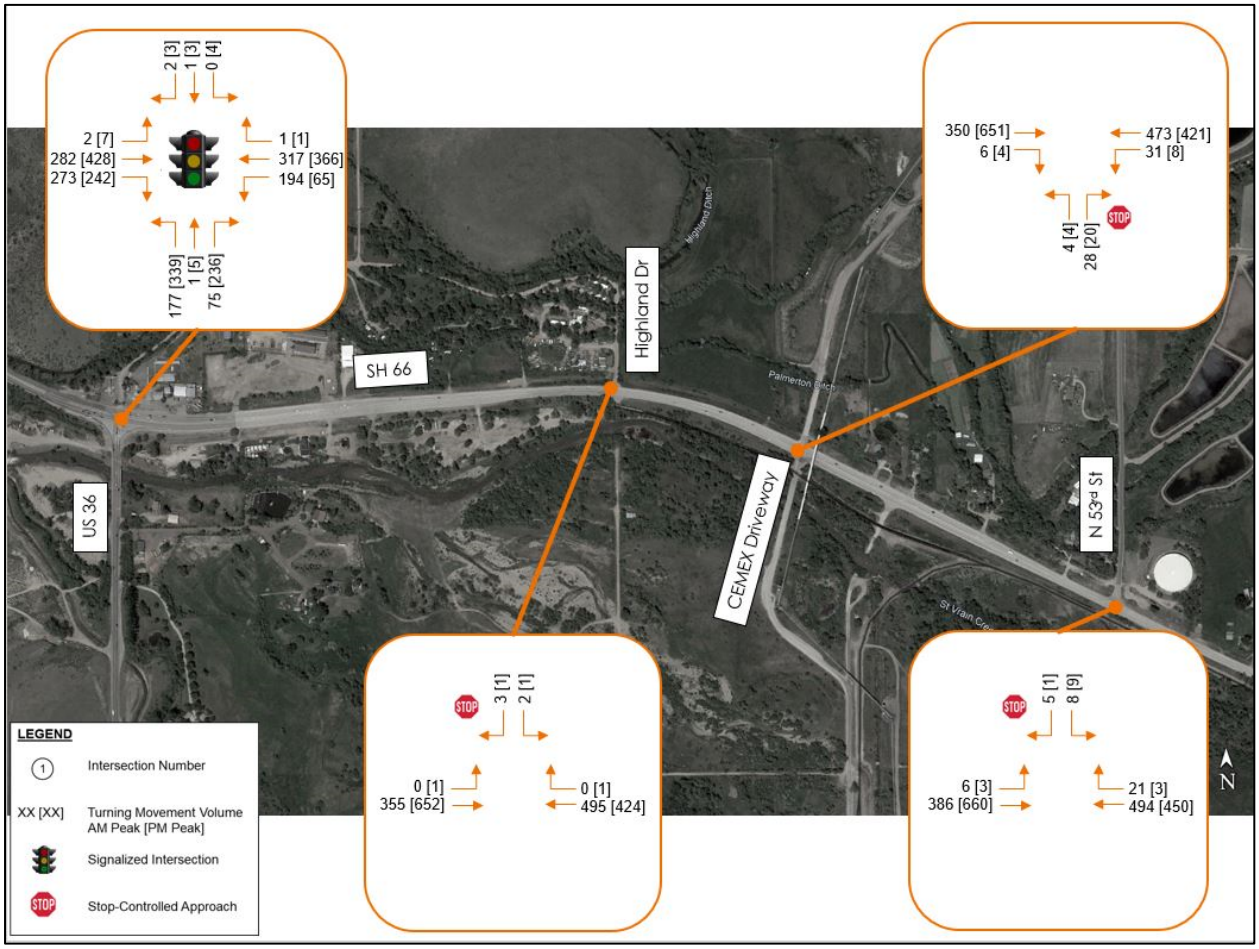
The scope of this traffic study included modeling existing, future background, and future total conditions in the study area, with the goal of completing an operational analysis of vehicular delay and Level of Service (LOS) at the study intersections. To account for varying daily traffic patterns, AM and PM peak period models were created for all conditions. Models were created using Synchro 11 software and analysis was completed using Highway Capacity Manual (HCM) methodologies, an industry-standard procedure used for evaluating traffic operations.

4.1 EXISTING CONDITIONS

Existing synchro models were developed using roadway geometrics and traffic control, traffic data, and growth rates as described within this report. For the primary purpose of analyzing conditions before the Dowe Flats Quarry closure, the year 2022 represents existing conditions. Results of these models present a baseline for comparing future operations and recommendations throughout the study.

A compound annual growth rate (CAGR) of 1.5% was extrapolated from the DRCOG 2020 and 2050 Model Assigned All-Day Volumes. Using this growth rate, negative growth was applied to the 2023 turning movement counts for all trips not directly associated with the CEMEX cement plant. Trips not associated with the CEMEX cement plant included those which did not enter or exit the plant during the observed peak hours. Estimated 2022 CEMEX generated trips, as described in **Section 3.2**, were added to the estimated 2022 non-CEMEX trips to develop the 2022 Existing traffic volumes shown in **Figure 7**.

Figure 7. 2022 Turning Movement Counts (Estimated)



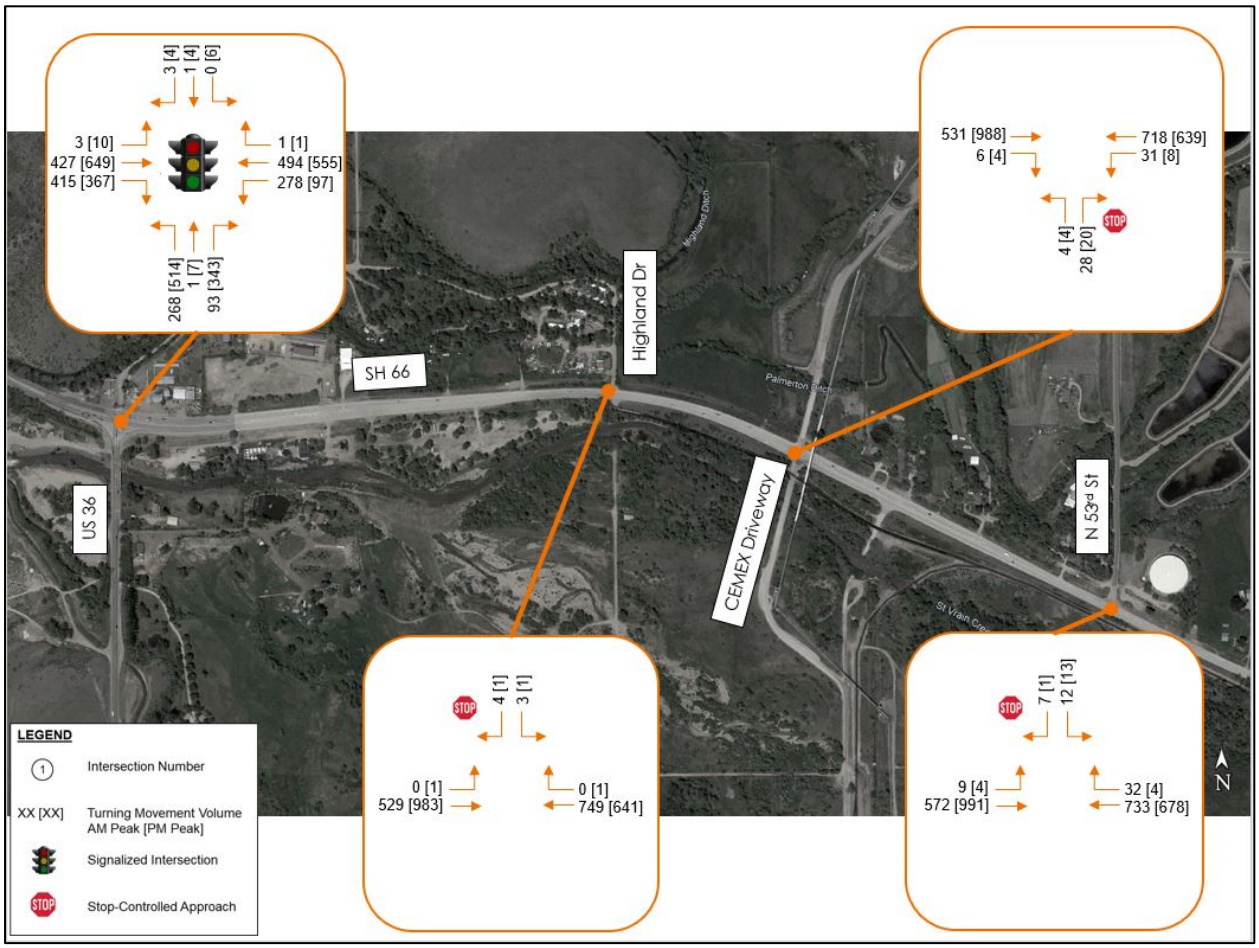
* All volumes reported as passenger car equivalent (PCE)

4.2 FUTURE CONDITIONS

Future conditions were considered as part of this analysis in order to quantify long-term impacts to the study area. To conform with DRCOG travel demand models, the year 2050 was used for the future horizon year. Existing Synchro models were modified to reflect Future Background and Future Total conditions. Future Background models represent conditions with no increase in truck trips at the site, while the Future Total models represent conditions in which there is an increase in truck trips due to the closure of the quarry. For the Future Background condition, the CAGR of 1.5% was applied to the 2023 turning movement counts and projected to the year 2050. Growth was only applied to vehicles not associated with CEMEX trip generation, as it is not expected that there will be an increase in material quantities or number of workers. Future Total models used this same methodology while also adding in

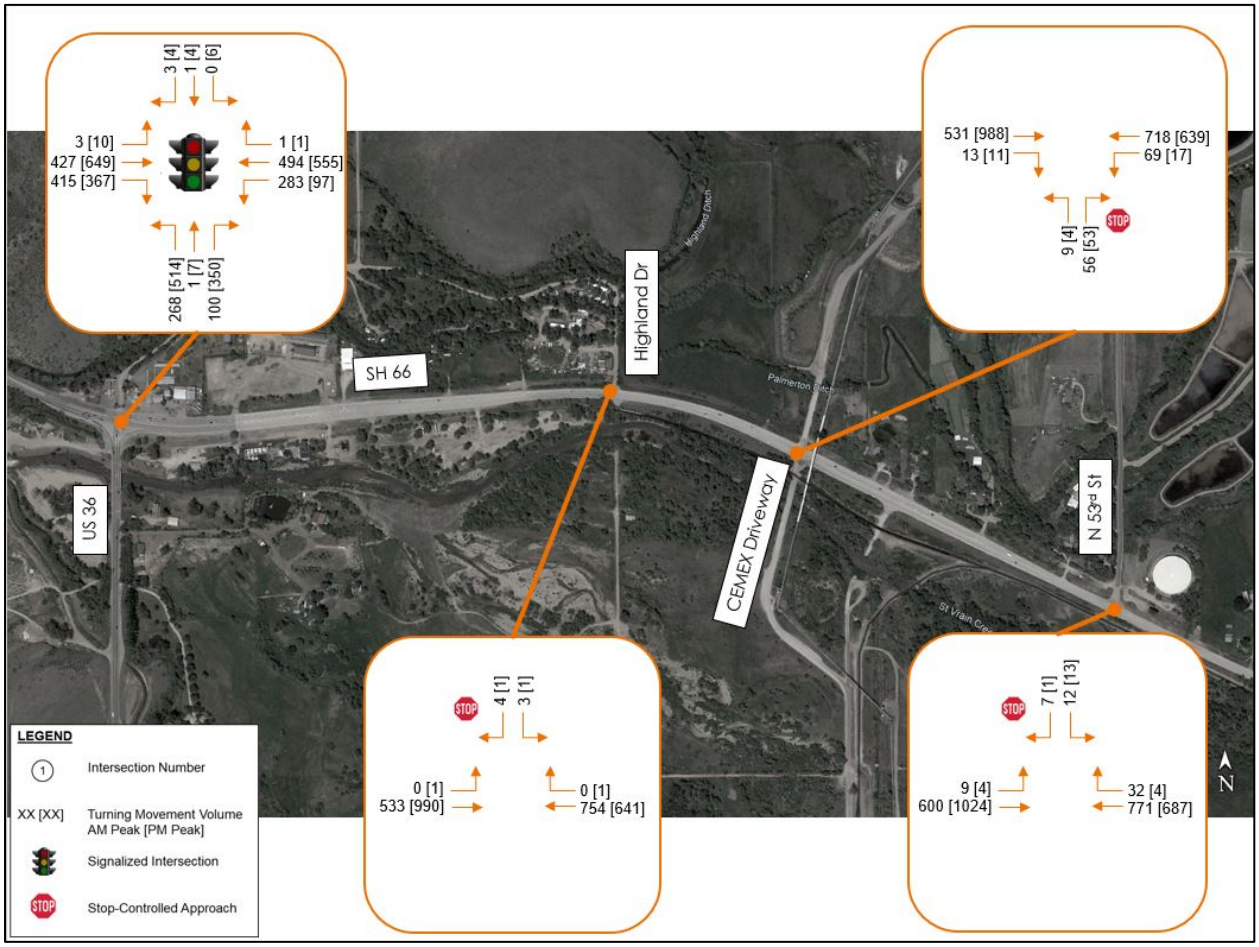
the additional limestone and shale truck trips that resulted from the Dowe Flats Quarry closure. Volumes for each of these future scenarios are displayed in **Figure 8** and **Figure 9**.

Figure 8. 2050 Future Background Intersection Volumes



* All volumes reported as passenger car equivalent (PCE)

Figure 9. 2050 Future Total Intersection Volumes



* All volumes reported as passenger car equivalent (PCE)

5.0 DELAY AND LEVEL OF SERVICE ANALYSIS

Using Synchro 11 traffic modeling software, reports were generated using HCM methodologies for each intersection included in this study. Average vehicle delay and LOS were used when determining how a given intersection is performing.

Table 4 displays the relationship between average vehicle delay and the corresponding LOS for both signalized and unsignalized movements. Typically, in rural areas such as this, LOS C or better is considered acceptable, while LOS D or worse may be considered unacceptable. A color spectrum has been included in the results tables in the following sections to help illustrate acceptable and unacceptable intersection conditions.

Table 4: Highway Capacity Manual Level of Service Criteria

LOS	Avg. Vehicle Delay (sec/veh)	
	Signalized Movement	Unsignalized Movement
A	0 – 10	0 – 10
B	10 – 20	10 – 15
C	20 – 35	15 – 25
D	35 – 55	25 – 35
E	55 – 80	35 – 50
F	80+	50+

It must be noted that, at unsignalized intersections that have free-flowing movements (i.e., two-way stop-controlled (TWSC) intersections), it is not valid to report LOS for movements that are free-flowing. This condition also applies to the intersection overall. In the results tables that follow, values for average delay have been provided for such free-flowing approaches, but LOS values have not.

The tables in the following sections present an overview of the delay and LOS results for each scenario. Detailed results reports generated in Synchro are provided in **Appendix C**.

5.1 2022 EXISTING CONDITIONS

The 2022 Existing Conditions scenario represents traffic prior to closure of the Dowe Flats Quarry. AM and PM Synchro models utilize existing traffic counts, geometry, and traffic control. Delay and LOS results of these models were used as a baseline for all future scenarios and are shown in **Table 5**.

Results for all intersections included within the study area are considered acceptable. While the southbound approach of US 36 and SH 66 is borderline acceptable at LOS D during the PM peak hour, the intersection, as a whole is operating acceptably at LOS C.

Table 5. 2022 Existing Synchro Results Summary

ID	Intersection	Approach	AM Peak			PM Peak		
			Volume (veh/hr)*	Delay (sec/veh)	LOS	Volume (veh/hr)*	Delay (sec/veh)	LOS
1	US 36 & SH 66	EB	557	18.8	B	677	22.9	C
		WB	512	9.0	A	432	14.0	B
		NB	241	21.5	C	571	23.1	C
		SB	3	30.7	C	10	36.9	D
		Total	**1313**	**15.7**	**B**	**1690**	**20.9**	**C**
2	SH 66 & Highland Dr	EB	352	0	-	650	0	-
		WB	495	0	-	425	0	-
		SB	5	14.1	B	2	16.8	C
		Total	**852**	**0.2**	**-**	**1077**	**0**	**-**
3	SH 66 & CEMEX Driveway	EB	356	0	-	655	0	-
		WB	504	0.5	-	429	0.2	-
		NB	32	12.4	B	24	17.3	C
		Total	**892**	**0.8**	**-**	**1108**	**0.7**	**-**
4	SH 66 & N 53rd St	EB	392	0.1	-	663	0	-
		WB	515	0	-	453	0	-
		SB	13	16.5	C	10	23.7	C
		Total	**920**	**0.3**	**-**	**1126**	**0.3**	**-**

* All volumes reported as passenger car equivalent (PCE)

5.2 2050 FUTURE BACKGROUND CONDITIONS

The 2050 Future Background Conditions combine 28 years of background growth with the number of CEMEX trips observed prior to the Dowe Flats Quarry closure.

Delay and LOS results, summarized in **Table 6**, show that all intersections operate at acceptable LOS during this scenario. All intersections exhibit unacceptable LOS for at least one approach during either the AM or PM peak period. For two-way stop-controlled intersections, failing LOS occurs along the stop-controlled approach. The southbound approach of SH 66 & N 53rd St is operating at LOS F. It should be noted that approaches exhibiting unacceptable LOS in this scenario is due to background growth and is not due to an increase in CEMEX truck traffic as a result of the Dowe Flats Quarry closure.



Table 6. 2050 Future Background Synchro Results Summary

ID	Intersection	Approach	AM Peak			PM Peak		
			Volume (veh/hr)*	Delay (sec/veh)	LOS	Volume (veh/hr)*	Delay (sec/veh)	LOS
1	US 36 & SH 66	EB	845	22.1	C	1026	30.9	C
		WB	773	10.7	B	653	19.1	B
		NB	362	26.2	C	864	38.9	D
		SB	4	37.4	D	14	44.1	D
		Total	**1984**	**18.7**	**B**	**2557**	**31**	**C**
2	SH 66 & Highland Dr	EB	529	0	-	984	0	-
		WB	749	0	-	642	0	-
		SB	7	21.9	C	2	30.3	D
		Total	**1285**	**0.3**	**-**	**1628**	**0**	**-**
3	SH 66 & CEMEX Driveway	EB	537	0	-	992	0	-
		WB	749	0.4	-	647	0.1	-
		NB	32	16.9	C	24	34.1	D
		Total	**1318**	**0.7**	**-**	**1663**	**0.8**	**-**
4	SH 66 & N 53rd St	EB	581	0.1	-	995	0	-
		WB	765	0	-	682	0	-
		SB	19	28.6	D	14	60.1	F
		Total	**1365**	**0.5**	**-**	**1691**	**0.6**	**-**

* All volumes reported as passenger car equivalent (PCE)

5.3 2050 FUTURE TOTAL CONDITIONS

Similar to the 2050 Future Background scenario, the 2050 Future Total scenario utilizes background growth for traffic not associated with the CEMEX cement plant. Additional truck trips, converted to PCE, are also included in this scenario to account for closure of the Dowe Flats Quarry. Results of these conditions remain relatively consistent with 2050 Future Background conditions, as displayed in **Table 7**.

The northbound approach of the SH 66 & CEMEX Driveway intersection operates at LOS E, due to the increase in through trips on SH 66 combined with the increase in traffic exiting the cement plant. Additionally, the southbound approach of the SH 66 & N 53rd St intersection operates at LOS F. Regardless of the approaches experiencing unacceptable LOS, described in this section, total intersection delays and LOS are considered acceptable.



Table 7. 2050 Future Total Synchro Results Summary

ID	Intersection	Approach	AM Peak			PM Peak		
			Volume (veh/hr)*	Delay (sec/veh)	LOS	Volume (veh/hr)*	Delay (sec/veh)	LOS
1	US 36 & SH 66	EB	845	22.3	C	1026	30.9	C
		WB	778	10.9	B	653	19.1	B
		NB	369	26.2	C	871	39.6	D
		SB	4	37.6	D	14	44.1	D
		Total	**1996**	**18.8**	**B**	**2564**	**31.2**	**C**
2	SH 66 & Highland Dr	EB	533	0	-	991	0	-
		WB	754	0	-	642	0	-
		SB	7	22	C	2	30.5	D
		Total	**1294**	**0.3**	**-**	**1635**	**0**	**-**
3	SH 66 & CEMEX Driveway	EB	544	0	-	999	0	-
		WB	787	0.8	-	656	0.3	-
		NB	65	21.5	C	57	37.6	E
		Total	**1396**	**1.7**	**-**	**1712**	**2.1**	**-**
4	SH 66 & N 53rd St	EB	609	0.1	-	1028	0	-
		WB	803	0	-	691	0	-
		SB	19	31.5	D	14	64.6	F
		Total	**1431**	**0.5**	**-**	**1733**	**0.6**	**-**

* All volumes reported as passenger car equivalent (PCE)

5.4 MITIGATION

As part of the scope of this study, mitigation would be considered for locations exhibiting failure in the 2050 Future Total scenario but not in the 2022 Existing or 2050 Future Background scenarios. Because the 2050 Future Total scenario represents the changing traffic conditions along the corridor, these are the results being considered.

While there are no locations exhibiting overall intersection failure in the 2050 Future Total scenario that were previously operating at acceptable levels, the northbound approach of SH 66 & CEMEX Driveway and the southbound approach of SH 66 & N 53rd St could be considered for mitigation. Due to the relatively low minor-street approach volumes, it is not anticipated that a signal warrant will be met for either location. It should be noted that excessive delays are only present along the northbound and southbound approaches and do not impact operations along SH 66. Recommendations are listed below.



SH 66 & CEMEX Driveway:

Primary Recommendation – Prohibit northbound left-turns out of the CEMEX cement plant. An alternate route is advised, utilizing SH 66, west of the cement plant. Minimally this should occur during peak traffic hours. An additional consideration is to limit hauling operations during AM and PM peak hours if truck operations present future issues exiting the cement plant.

Secondary Recommendation – Provide a northbound left-turn acceleration lane along SH 66. This can be accomplished by providing protected, downstream, left-turn storage which would allow for these movements to enter the major-street with less conflict. This acceleration lane would be protected from westbound through traffic via raised median. For safety reasons, the westbound through traffic would ultimately merge with the acceleration lane.

Per *AASHTO Geometric Design of Highways and Streets, 2011*, the acceleration lane should be a minimum of 1200 feet in length. This assumes that the turning vehicle is beginning at a stopped condition and will be combined with through traffic at the design speed of 60 MPH. Because trucks account for the majority of trips making the northbound left-turn movement, it is suggested that this length be increased to account for the slower acceleration rate of trucks.

SH 66 & N 53rd St:

Primary Recommendation – Provide a southbound left-turn acceleration lane along SH 66. This can be accomplished by providing protected, downstream, left-turn storage which would allow for these movements to enter the major-street with less conflict. This acceleration lane would be protected from eastbound through traffic via raised median. For safety reasons, the eastbound through traffic would ultimately merge with the acceleration lane.

Per *AASHTO Geometric Design of Highways and Streets, 2011*, the acceleration lane should be a minimum of 1200 feet in length. This assumes that the turning vehicle is beginning at a stopped condition and will be combined with through traffic at the design speed of 60 MPH.

6.0 CONCLUSIONS AND SUMMARY

The following conclusions and recommendations are summarized based on the analysis presented in this report:

For the sake of this analysis, estimated 2022 traffic volumes were considered for existing conditions. While this report was completed in 2023, closure of the Dowe Flats Quarry occurred in 2022 which resulted in an increase of daily truck traffic, entering and exiting the CEMEX cement plant. This increase was the primary focus of the study.

Using Synchro 11 software, 2022 Existing, 2050 Future Background, and 2050 Future Total conditions were modeled. Results of this analysis showed the following:

2022 Existing – This model represents conditions prior to the observed increase in truck traffic. All intersections operate at LOS C or better and with minimal delay. The southbound approach at the US 36 & SH 66 intersection operates at LOS D.

2050 Future Background – This model represents future traffic conditions, under the assumption that there was no increase in truck traffic. A compound annual growth rate of 1.5% was applied to all non-CEMEX trips. Intersections in this scenario continue to operate with acceptable delay and LOS. However, each intersection now demonstrates poor LOS for select approaches during the AM and PM peak periods. This is expected due to the significant increase of background traffic volumes expected over time.

2050 Future Total – This model represents future traffic conditions while considering the increase in truck trips entering and exiting the CEMEX cement plant. A compound annual growth rate of 1.5% was applied to all non-CEMEX trips. This scenario shows little change when compared to 2050 Future Background conditions. All intersections continue to operate acceptably, with specific approaches operating unacceptably for AM and PM peak periods. Delay and LOS gets slightly worse as a result of the increased number of trips generated by the CEMEX site.

Recommendations were made on a locational basis and provided two alternatives for the SH 66 & CEMEX Driveway intersection and one alternative for the SH 66 & N 53rd St intersection.

In order to alleviate excessive delay and LOS along minor street approaches, left-turn acceleration lanes can be provided to allow for turning vehicles to enter the traffic steam while only traversing traffic from the adjacent through movement. Raised medians can provide protection and lane assignment from adjacent, merging traffic. Acceleration lanes should be in accordance with design minimums described in *AASHTO Geometric Design of Highways and Streets, 2011*. Design lengths should also take into account the slower acceleration rate of trucks for the SH 66 and CEMEX Driveway intersection.

Another possible mitigation measure for the SH 66 & CEMEX Driveway intersection is to prohibit northbound left-turns and develop an alternate route along SH 66, west of the cement plant. Lastly, if permitted and necessary, a shift in hauling operations, outside of SH 66 AM and PM peak hours, should be considered.

Because it was deemed that unacceptable intersection results were not directly attributed to CEMEX cement plant operations, but rather from background growth, future mitigation should be considered and executed, as desired, by the owning entity. The failing northbound approach at the intersection of SH 66 & CEMEX Driveway is only a matter of inconvenience to CEMEX operations and has no negative impacts to traffic along SH 66. Similarly, the failing southbound approach at the intersection of SH 66 & N 53rd St should be addressed by CDOT as it is unrelated to the increase in CEMEX generated truck trips.